Golden Coordinates at Level Five: Ruled-Surface Rank Loss, CM Values, and Fourier Self-Duality

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Abstract

The interior rank loss of an orthogonal-circle ruled surface selects $c_0 = (3-\sqrt{5})/2 = \phi^{-2}$. We give explicit maps connecting this coordinate to level-five arithmetic and Fourier symmetry. At $\tau_{163} = (1+\mathrm{i}\sqrt{163})/2$, an exact factorization of Klein's equation recovers $\sqrt{5}$ and hence c_0 from $v = r(\tau_{163})^5$. A square witness identifies $\mathbb{Q}(v) = \mathbb{Q}(v - v^{-1})(\sqrt{163(1+c_0)})$; the radical is a pentagon side length at circumradius $\sqrt{163}$. At the distinct point i, Rogers–Ramanujan inversion is conjugate by coordinate exchange to the odd period-five sine-Fourier involution. Its two eigenlines characterize the two signed completed functional equations by Poisson–Mellin analysis. Exact arithmetic certificates are kept separate from the analytic arguments.

1 Introduction and principal assertions

Hermite's 1859 discussion leads, in modern notation, to the remarkably small gap between $\mathrm{e}^{\pi\sqrt{163}}$ and an integer [4, pp. 47–48]. For $\tau_{163} = (1+\mathrm{i}\sqrt{163})/2$ and $q = \mathrm{e}^{2\pi\mathrm{i}\tau_{163}} = -\mathrm{e}^{-\pi\sqrt{163}}$, the classical j-expansion $q^{-1} + 744 + 196884q + \cdots$ and $j(\tau_{163}) = -640320^3$ explain its small gap [3]. Meanwhile $P(c) = c^2 - 3c + 1$ has discriminant 5, so its roots lie in $\mathbb{Q}(\sqrt{5}) = \mathbb{Q}(\zeta_5 + \zeta_5^{-1})$. Indeed $\zeta_5 + \zeta_5^{-1} = 2\cos(2\pi/5) = (\sqrt{5}-1)/2$ has minimal polynomial $X^2 + X - 1$, identifying the real quadratic cyclotomic subfield. The aim is to identify maps that recover this scalar from a CM value and carry it into Fourier symmetry.

Throughout, put $\phi = (1+\sqrt{5})/2$, $\gamma = \phi^{-1}$, $c_0 = \phi^{-2} = \gamma^2$, and $u_0 = \gamma/2$. The arithmetic connection is explicit: Theorem 3.1 recovers $\sqrt{5}$ as a rational function of the actual CM value $v = r(\tau_{163})^5$, where r is the Rogers–Ramanujan function defined in (3). Corollary A.1 then places the length $\delta = \sqrt{163(1+c_0)} = 2\sqrt{163}\sin(\pi/5)$ inside $\mathbb{Q}(v)$. The companion length $\phi\delta$ is the diagonal of the same regular pentagon. Thus its squared side-to-diagonal ratio is precisely the surface coordinate c_0.

The second connection identifies two actions on coordinates. The matrices in Section 4 satisfy $A = JMJ$: normalized modular inversion becomes the odd five-periodic sine-Fourier involution after exchanging the two coordinates. Proposition 4.1 identifies its positive eigenline through $\kappa = r(i)$. Theorems 5.1 and 5.2 carry this finite symmetry to an entire completed function and characterize both signed functional equations. The CM value and modular inversion are classical inputs from Duke; the rational recovery, square witness and coordinate maps are explicit here, and the analytic step uses Poisson summation. Figure 1 separates the arithmetic role of τ_{163} from the fixed-point role of i.

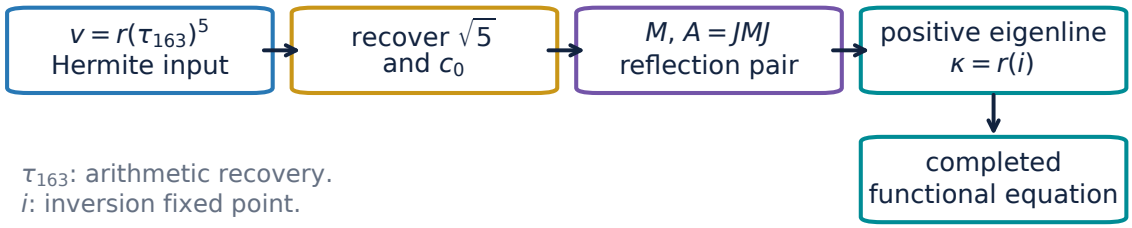


Figure 1: The two CM arguments enter at different stages. The value at τ_{163} recovers the golden scalar; the fixed point i selects the positive Fourier eigenline. The arrows summarize the explicit maps proved in the text.

2 The surface and its golden coordinate

Following the paired-circle construction [5, Definition 1.1], use $-\pi/2 < t < \pi/2$ and $0 \le u \le 1$. The surface angle is t and the ruling coordinate is u:

$$S(t,u) = (c + 2u(1-c), (1-u)\sin t, u\sqrt{c(2-c)}), \qquad c = \cos t. \tag{1}$$

The paired endpoints are $C_1(t) = (c, \sin t, 0)$ and $C_2(t) = (2-c, 0, \sqrt{c(2-c)})$: points on unit-circle arcs in perpendicular planes, with centers $(0,0,0)$ and $(1,0,0)$. Thus $S(t,u) = (1-u)C_1(t) + uC_2(t)$ for $0 \le u \le 1$. For fixed t, varying u traces a ruling segment. Rank loss means that its two parameter-direction tangent vectors become linearly dependent. In $0 < c < 1$, the third component of $S_t \times S_u$ is $(1-c)(1-c-2u)$, hence rank loss forces $u = (1-c)/2$. The remaining components are

$$-\frac{(1+c)P(c)}{2\sqrt{c(2-c)}}, \qquad -\frac{\sin t\, P(c)}{\sqrt{c(2-c)}}.$$

The prefactor of the first remaining component is nonzero in $0 < c < 1$. Consequently, in this interior range,

$$\operatorname{rank}(dS) < 2 \quad \Longleftrightarrow \quad u = (1-c)/2 \text{ and } P(c) = 0.$$

The unique root of P in $(0,1)$ therefore gives

$$c_0 = \frac{3-\sqrt{5}}{2}, \qquad \gamma = 1 - c_0 = \phi^{-1}, \qquad u_0 = \gamma/2 = \cos(2\pi/5). \tag{2}$$

The corresponding parameter points are $(t,u) = (\pm \arccos c_0, u_0)$; we call them the reflected lateral pair. Figure 2 displays the surface and the two conditions for rank loss. This local calculation is also the interior part of the published five-singularity classification [5, Theorem 2.2].

The projection $u_0 = \cos 72^\circ$ belongs to a unit pentagon, whose squared side-to-diagonal ratio is c_0. A second Euclidean realization uses the unit-radius icosahedron with vertices

$$\frac{(0, \pm 1, \pm\phi),\ (\pm 1, \pm\phi, 0),\ (\pm\phi, 0, \pm 1)}{\sqrt{1+\phi^2}}.$$

Its distinct dot products are $1/\sqrt{5}, -1/\sqrt{5}, -1$, with pair counts $30, 30, 6$. The squared short-to-long chord ratio is $(2 - 2/\sqrt{5})/(2 + 2/\sqrt{5}) = c_0$. Supplementary plates S04 and S12 show these models; the full surface and parameter diagnostics are in S02–S03.

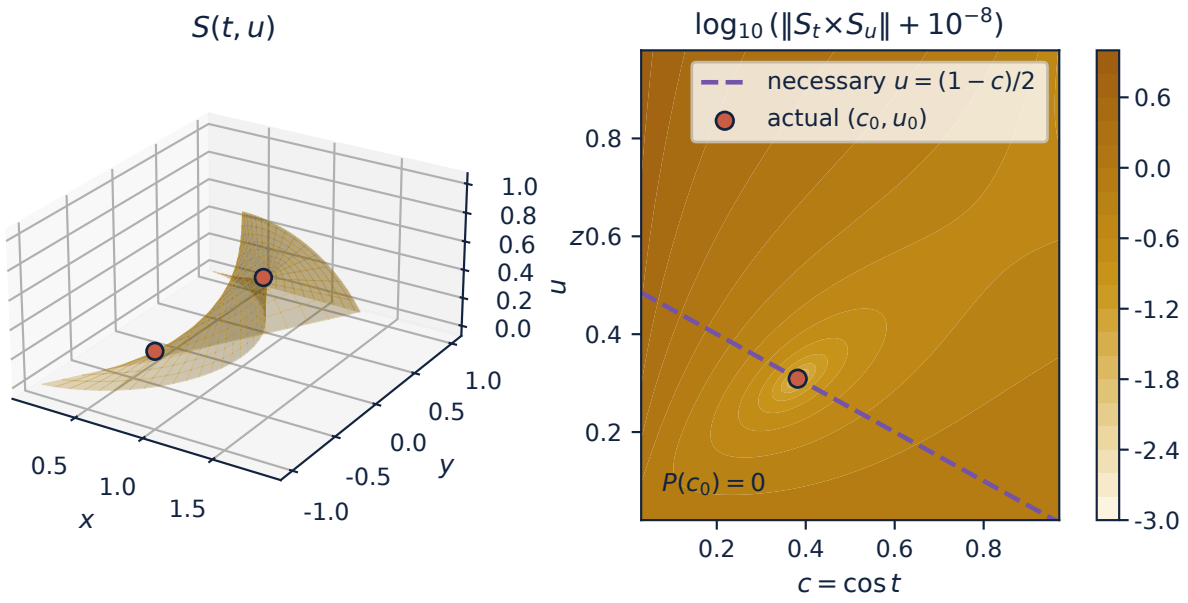


Figure 2: The actual ruled patch and its rank-loss diagnostic. On the positive-sine parameter branch, the dashed line $u = (1 - c)/2$ is necessary; rank loss occurs at its marked point (c_0, u_0), where $P(c_0) = 0$. The color scale is $\log_{10}(\|S_t \times S_u\| + 10^{-8})$. The surface panel marks both reflected parameter points.

3 CM field recovery

For $\Im\tau > 0$, use the Rogers–Ramanujan product [3, Eq. (4.1)]

$$r(\tau) = q^{1/5} \prod_{n\geq 1} \frac{(1 - q^{5n-4})(1 - q^{5n-1})}{(1 - q^{5n-3})(1 - q^{5n-2})}, \quad q = \mathrm{e}^{2\pi \mathrm{i}\tau}, \quad q^{1/5} = \mathrm{e}^{2\pi \mathrm{i}\tau/5}. \tag{3}$$

The two arguments have the following roles:

Argument	Nome q	j-value	Role
i	$\mathrm{e}^{-2\pi}$	1728	inversion fixed point
τ_{163}	$-\mathrm{e}^{-\pi\sqrt{163}}$	-640320^3	arithmetic recovery

Set $v = r(\tau_{163})^5$. The classical CM value above and Klein's level-five relation (as recorded by Duke [3, Eq. (2.5)]) give

$$P_j(v) = (v^4 - 228v^3 + 494v^2 + 228v + 1)^3 + jv(v^2 + 11v - 1)^5 = 0. \tag{4}$$

Let $A_{163}, B_{163} \in \mathbb{Z}[X]$ be the explicit polynomials in Appendix A.

Theorem 3.1. *For $v = r(\tau_{163})^5$, $\sqrt{5} \in \mathbb{Q}(v)$ and*

$$\sqrt{5} = -\frac{A_{163}(v)}{B_{163}(v)}, \qquad \phi = \frac{B_{163}(v) - A_{163}(v)}{2B_{163}(v)},$$
$$\gamma = -\frac{A_{163}(v) + B_{163}(v)}{2B_{163}(v)}, \quad c_0 = \frac{3B_{163}(v) + A_{163}(v)}{2B_{163}(v)}.$$

Proof. Appendix A gives the exact identity $P_{-640320^3}(X) = A_{163}(X)^2 - 5B_{163}(X)^2$ with coprime integer polynomials. If $B_{163}(v) = 0$, then $A_{163}(v) = 0$, contrary to Bezout. Thus $(A_{163}(v)/B_{163}(v))^2 = 5$. The rational interval certificate gives $A_{163}(v) < 0 < B_{163}(v)$, selecting $-\sqrt{5}$; the formulas follow. □

The square witness in Appendix A strengthens this recovery: with $U = v - v^{-1}$, it proves

$$\mathbb{Q}(v) = \mathbb{Q}(U)(\delta), \qquad \delta = 2\sqrt{163}\sin(\pi/5).$$

The pentagonal side length is therefore represented inside the field of the selected CM value by an explicit formula, not just by its defining radical.

For the near-integer motivation, put $R = e^{\pi\sqrt{163}}$ and $N = 640320^3+744 = 262537412640768744$. Substituting $q = -1/R$ in the classical j-series gives

$$N - R = \frac{196884}{R} - \frac{21493760}{R^2} + \frac{864299970}{R^3} + \cdots \approx 7.499274028 \times 10^{-13}.$$

For these coefficients, $j = E_4^3/\Delta$, where $E_4(q) = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n$, $\Delta(q) = q\prod_{n\geq 1}(1-q^n)^{24}$, and $\sigma_3(n) = \sum_{d|n} d^3$. Figure 3 shows successive corrections to this one fixed gap. The complete CM comparison and numerical ladder are in S10–S11.

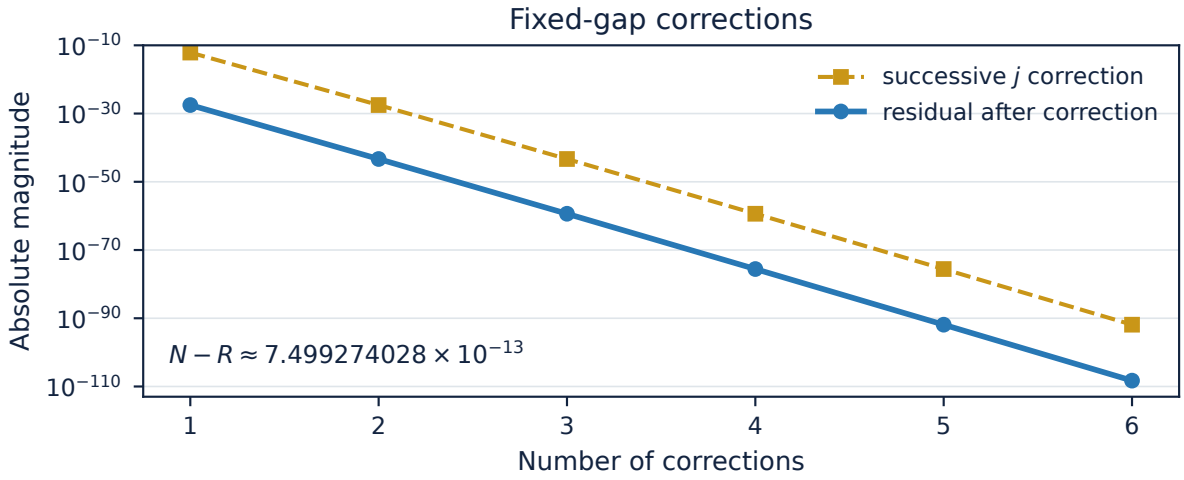


Figure 3: Successive j-series corrections to the fixed gap $N-R$. For correction count m, the residual is $|N - R - \sum_{n=1}^{m}(-1)^{n+1}c_n/R^n|$, where $c_1 = 196884$, $c_2 = 21493760$, $c_3 = 864299970$. Values use 180 decimal digits and six corrections; the series coefficients are obtained by exact expansion of E_4^3/Δ.

4 The common involution

Duke's inversion law [3, Proposition 2, Eq. (3.2)] is

$$r(-1/\tau) = \frac{\gamma - r(\tau)}{1 + \gamma r(\tau)}.$$

In homogeneous coordinates put

$$M = \frac{1}{\sqrt{1+c_0}}\begin{pmatrix} -1 & \gamma \\ \gamma & 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Taking the ratio of $M(z,1)^T$ gives the displayed law, and direct multiplication, using $\gamma^2 = c_0$, proves $M^2 = I$.

Let V_5 be the real space of odd five-periodic sequences. In residues $1,2,3,4,5$ (with $5 = 0$), it consists of $(x, y, -y, -x, 0)$. Define

$$(\mathcal{F}a)_m = \frac{1}{\sqrt{5}} \sum_{n=1}^{5} a_n \mathrm{e}^{2\pi \mathrm{i} mn/5}, \qquad \mathcal{T} = -\mathrm{i}\mathcal{F}.$$

Pairing n and $5-n$ gives

$$A = \frac{2}{\sqrt{5}} \begin{pmatrix} \sin(2\pi/5) & \sin(\pi/5) \\ \sin(\pi/5) & -\sin(2\pi/5) \end{pmatrix} = JMJ. \tag{5}$$

Indeed $2\sin(2\pi/5)/\sqrt{5} = 1/\sqrt{1+c_0}$ and $2\sin(\pi/5)/\sqrt{5} = \gamma/\sqrt{1+c_0}$, by squared pentagon identities. Thus $A^2 = I$.

Proposition 4.1. *With $\kappa = r(i)$,*

$$\kappa = \frac{\gamma}{1+\sqrt{1+c_0}}, \qquad A(1,\kappa)^T = (1,\kappa)^T.$$

For $\theta = \frac{1}{2}\arctan\gamma$,

$$\kappa = \tan\theta, \quad \cos 2\theta = \frac{1}{\sqrt{1+c_0}}, \quad \sin 2\theta = \frac{\gamma}{\sqrt{1+c_0}},$$

and, separately, $\kappa = 2\cos 18^\circ - 2\cos 36^\circ$. Finally $u_0 = \kappa/(1-\kappa^2)$ and $c_0 = (2\kappa/(1-\kappa^2))^2$.

Proof. The inversion law fixes $r(i)$; positivity of every product factor in (3) at i selects the positive root of $\gamma X^2 + 2X - \gamma = 0$. It gives the stated κ and $M(\kappa,1)^T = (\kappa,1)^T$, so J transfers it explicitly to the $+1$ eigenvector of A. Since $\tan 2\theta = \gamma$ and $0 < 2\theta < \pi/2$, right-triangle and half-angle formulas prove the three half-angle identities. This axis angle is not 18°. Separately $2\cos 36^\circ = \phi$, $2\cos 18^\circ = \sqrt{2+\phi}$; squaring positive sides shows their difference has the same positive quadratic root. The last formulas are the tangent double-angle identity. □

The fixed-point polynomial also has a norm interpretation:

$$X^4 + 2X^3 - 6X^2 - 2X + 1 = (X^2 + 2\phi X - 1)(X^2 - 2\phi^{-1}X - 1),$$

the polynomial norm $N_{\mathbb{Q}(\sqrt{5})(X)/\mathbb{Q}(X)}(X^2 + 2\phi X - 1)$. Conjugation sends ϕ to $-\phi^{-1}$; multiplication using $\phi - \phi^{-1} = 1$ and $\phi\phi^{-1} = 1$ proves the identity. The fixed-point equation gives $\kappa^2 + 2\phi\kappa - 1 = 0$, so the first factor vanishes at $X = \kappa$.

Figure 4 shows the coordinate exchange and the positive Fourier coefficient pattern. Supplementary plates S05–S07 give the full reflection, direct-product and finite-DFT diagnostics.

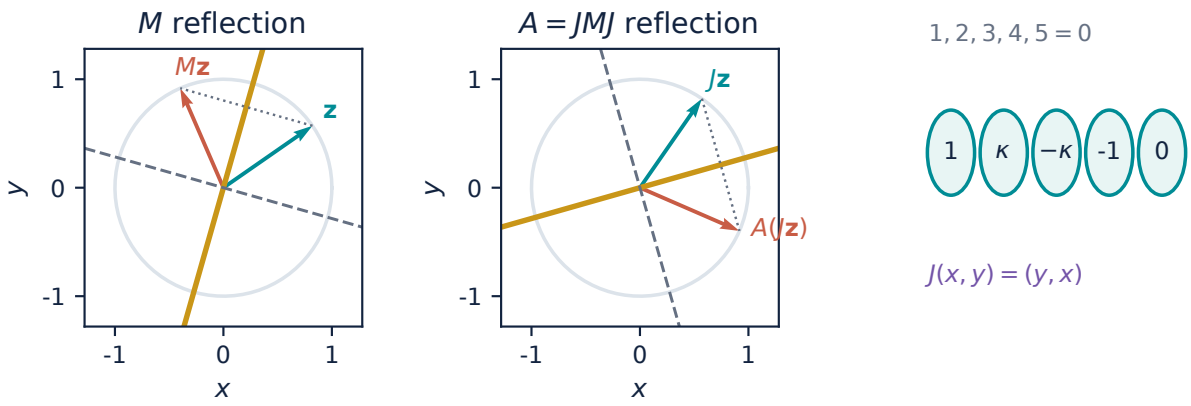


Figure 4: Coordinate exchange between the reflections M and $A = JMJ$. The paired off-axis inputs are $\mathbf{z}$ and $J\mathbf{z}$; their images satisfy $A(J\mathbf{z}) = J(M\mathbf{z})$. The solid axes are the positive eigenlines, with directions $(\kappa, 1)$ and $(1, \kappa)$ respectively; the dashed axes have eigenvalue -1. The residue strip gives the normalized positive Fourier eigenvector in order $1, 2, 3, 4, 5 = 0$.

5 Poisson–Mellin bridge and classification

The Gaussian bridge turns finite Fourier symmetry into analytic symmetry. To retain t as a surface angle, use $x > 0$ here:

$$F_a(s) = \sum_{n\geq 1} a_n n^{-s} \quad (\Re s > 1), \qquad \Theta_a(x) = \sum_{n\in\mathbb{Z}} n a_n \mathrm{e}^{-\pi n^2 x/5}.$$

Theorem 5.1. *For $a \in V_5$, $a' = \mathcal{T}a$,*

$$\Lambda_a(s) = \frac{1}{2}\int_0^\infty \Theta_a(x) x^{(s+1)/2}\frac{dx}{x} = \left(\frac{5}{\pi}\right)^{(s+1)/2}\Gamma((s+1)/2)F_a(s)$$

defines an entire function by its integral; the displayed Dirichlet-series expression holds for $\Re s > 1$. Moreover $\Lambda_a(s) = \Lambda_{a'}(1-s)$.

Proof. With $\widehat{f}(\xi) = \int_{\mathbb{R}} f(y)\mathrm{e}^{-2\pi \mathrm{i} y\xi} dy$, put $f_x(y) = y\mathrm{e}^{-\pi x y^2/5}$. Its transform is

$$\widehat{f}_x(\xi) = -\mathrm{i}(5/x)^{3/2}\xi \mathrm{e}^{-5\pi\xi^2/x}.$$

Residue-class Poisson summation [2, §1.8(iv)] is

$$\sum_{\ell\in\mathbb{Z}} f_x(j+5\ell) = \frac{1}{5}\sum_{k\in\mathbb{Z}} \mathrm{e}^{2\pi \mathrm{i} kj/5}\widehat{f}_x(k/5).$$

After multiplying by a_j and summing,

$$\begin{aligned}\Theta_a(x) &= \frac{1}{\sqrt{5}}\sum_k (\mathcal{F}a)_k \widehat{f}_x(k/5)\\ &= \frac{1}{\sqrt{5}}\sum_k \mathrm{i}a'_k[-\mathrm{i}(5/x)^{3/2}(k/5)\mathrm{e}^{-\pi k^2/(5x)}] = x^{-3/2}\Theta_{a'}(1/x).\end{aligned}$$

This shows the $1/5$, $1/\sqrt{5}$, $\mathrm{i}(-\mathrm{i})$ cancellation, and $x^{-3/2}$. Bounded periodicity gives exponential decay at infinity; the transformation gives $Cx^{-3/2}\mathrm{e}^{-c/x}$ at zero. These envelopes, also after multiplication by $|\log x|^r$, prove local uniform convergence of the integral and all s derivatives. For $\Re s > 1$, odd pairing and termwise gamma integration give the displayed completion. Substitution of the theta transformation changes $x^{(s+1)/2}$ to $x^{(s-2)/2}$, and $y = 1/x$ gives $y^{(2-s)/2}dy/y$, proving the functional equation. □

Theorem 5.2. *For nonzero $a \in V_5$ and $\epsilon \in \{+1, -1\}$, $\Lambda_a(s) = \epsilon\Lambda_a(1-s)$ for all s iff $\mathcal{T}a = \epsilon a$. The positive line is $\mathbb{R}(1, \kappa, -\kappa, -1, 0)$, fixed by $a_1 = 1$; the negative line is $\mathbb{R}(1, -1/\kappa, 1/\kappa, -1, 0)$, and its completed value at $1/2$ is zero.*

Proof. Root-of-unity orthogonality gives $\mathcal{F}^2 a = a_{-n} = -a$, hence $\mathcal{T}^2 = I$. Solving the two eigenline equations of (5) gives the stated lines. One implication follows from Theorem 5.1. Conversely let $b = \mathcal{T}a - \epsilon a$. The assumed equation and that theorem make F_b vanish in a right half-plane. If m is its first nonzero coefficient, then for real $\sigma \to \infty$,

$$0 = m^\sigma F_b(\sigma) = b_m + \sum_{n>m} b_n(m/n)^\sigma.$$

The bounded tail tends to zero because

$$\left|\sum_{n>m} b_n(m/n)^\sigma\right| \le Cm^\sigma \int_m^\infty x^{-\sigma}dx = \frac{Cm}{\sigma - 1}.$$

Thus $b_m = 0$, a contradiction. Central vanishing is immediate for $\epsilon = -1$. □

For a general vector, put $a_\pm = (a \pm \mathcal{T}a)/2$. Its completion is the sum of a symmetric and an antisymmetric part. If both components are nonzero, neither scalar-sign equation holds. For example, the vector with $(x, y) = (1, 0)$ is an eigenvector of neither sign. The normalized $+1$ line is selected by $a_1 = 1$ and gives the Davenport–Heilbronn sequence; its parameter is the classical constant $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1)$.

Here $a_6 = 1$ but $a_2 a_3 = -\kappa^2$, so the normalized $+1$ sequence is nonmultiplicative. The off-critical-line zero result is classical and externally attributed [1]; this is only a classification inside V_5.

For the normalized positive vector $a = (1, \kappa, -\kappa, -1, 0)$, the paired theta transformation becomes $\Theta_a(x) = x^{-3/2}\Theta_a(1/x)$. Supplementary plates S08–S09 show independent numerical evaluations of this identity and the completed function. On a fixed-height complex slice, evenness of the real part and oddness of the imaginary part about $\Re s = 1/2$ use both the functional equation and $\Lambda_a(\bar{s}) = \overline{\Lambda_a(s)}$, since the coefficients are real. The numerical methods and saved data are stated in the companion.

6 Conclusion

The surface calculation supplies the golden coordinate, the CM certificate recovers it from the actual τ_{163} value, and $A = JMJ$ supplies the analytic payoff at i. The recovered golden scalar determines a reflection, whose positive eigendirection determines the analytic symmetry. The explicit maps connect the geometric ratio, arithmetic field and functional equation through transformations that can each be checked. A precise remaining question is whether an N-dependent ruled-surface family can be constructed to carry the odd-N Fourier mechanism below.

Associated files. The accompanying source bundle contains the article, the visual companion (plates S01–S12), the plotting data and scripts, and exact verification programs. Its README gives replay commands and file hashes. The geometric source used in [5] is included with its published-source hash.

A Exact certificate and square witness

Put $U = v - v^{-1}$ and

$$\alpha(U) = U^3 - p_1U^2 + (3 - p_2)U - (2p_1 + p_3),$$
$$\beta(U) = -d_1U^2 - d_2U - (2d_1 + d_3).$$

The integer coefficients are

$$\begin{aligned} p_1 &= 131268706320384342, & d_1 &= 58705150130167150,\\ p_2 &= 2896632145572621515, & d_2 &= 1295413276662289800,\\ p_3 &= 15717124384666951860, & d_3 &= 7028911706986972100. \end{aligned}$$

The reconstruction polynomials are

$$A_{163}(X) = X^6 - p_1X^5 - p_2X^4 - p_3X^3 + p_2X^2 - p_1X - 1,$$
$$B_{163}(X) = -d_1X^5 - d_2X^4 - d_3X^3 + d_2X^2 - d_1X.$$

Exact expansion gives $P_{-640320^3}(X) = A_{163}(X)^2 - 5B_{163}(X)^2$ and $\gcd(A_{163}, B_{163}) = 1$, equivalently $P_{-640320^3}(v) = v^6(\alpha(U)^2 - 5\beta(U)^2)$.

Define $G(q)$ as the positive product in (3), so $v = qG(q)^5$. For $q = -\mathrm{e}^{-\pi\sqrt{163}}$ every factor is positive and the phase is $r(\tau_{163}) = \mathrm{e}^{\pi\mathrm{i}/5}(-v)^{1/5}$. The rational estimates

$$3.14 < \pi < 22/7, \quad 12.76 < \sqrt{163} < 12.77, \quad \mathrm{e}^{40} > 23 \cdot 10^{16}, \quad \mathrm{e}^{41} < 7 \cdot 10^{17},$$

$$|\log G(q)| \le \frac{|q|}{(1 - |q|)^2}$$

give $40 < \pi\sqrt{163} < 41$, $5|\log G(q)| < 1/1000$, and

$$-5 \cdot 10^{-18} < v < -10^{-18}.$$

Exact interval Horner arithmetic gives $A_{163}(v) < 0 < B_{163}(v)$ throughout this rational interval; a Sturm count gives exactly one root of $P_{-640320^3}(X)$ there. The source bundle replays these exact finite computations and records separate 100/200-digit product diagnostics.

With $S=\sqrt{5}$, define

$$\begin{aligned} w(U)=\frac{1}{1574580000}[&(312119004989-139583862445S)U^2\\ &+(6858643496508-3067324783740S)U\\ &+37679427598544-16850735555920S]. \end{aligned}$$

Exact reduction by $\alpha(U)+S\beta(U)$ and S^2-5 proves

$$U^2+4=163(1+c_0)w(U)^2.$$

On the selected branch $U>1$, the leading and linear coefficients are negative and $w(1)<0$, hence $w'(U)<0$ and $w(U)<0$.

Corollary A.1. *For $\delta=\sqrt{163(1+c_0)}>0$, $\mathbb{Q}(v)=\mathbb{Q}(U)(\delta)$.*

Proof. The coefficient formulas also give $A_{163}(v)=v^3\alpha(U)$ and $B_{163}(v)=v^3\beta(U)$. Hence $S=-\alpha(U)/\beta(U)\in\mathbb{Q}(U)$ and $w(U)\in\mathbb{Q}(U)$. Since $U\in\mathbb{Q}(v)$, both also belong to $\mathbb{Q}(v)$. Since $2v-U=v+v^{-1}<0$ and $w(U)<0$, $\delta=(2v-U)/w(U)\in\mathbb{Q}(v)$. Conversely $v=(U+\delta w(U))/2\in\mathbb{Q}(U)(\delta)$. □

The square-witness verifier in the source bundle replays the zero remainder and the rational sign comparisons. The selected radical has a direct pentagonal interpretation:

$$\delta=2\sqrt{163}\sin(\pi/5),\qquad \phi\delta=2\sqrt{163}\sin(2\pi/5).$$

Indeed $1+c_0=(5-\sqrt{5})/2=4\sin^2(\pi/5)$ and $2\cos(\pi/5)=\phi$. The two lengths are the side and diagonal of a regular pentagon of circumradius $\sqrt{163}$; their squared ratio is c_0. Eliminating S from $\delta^2=163(5-S)/2$ gives $\delta^4-815\delta^2+132845=0$, a different polynomial from the norm quartic for κ. The cyclic quartic field of conductor 815 is a known family [6]; the polynomial here records the selected generator used above.

B Odd-N extension

For odd $N>1$, let V_N be real odd N-periodic sequences and define

$$(\mathcal{F}_N a)_m=N^{-1/2}\sum_{n=1}^{N}a_n\mathrm{e}^{2\pi\mathrm{i}mn/N},\qquad \mathcal{T}_N=-\mathrm{i}\mathcal{F}_N.$$

Its real sine matrix is

$$\frac{2}{\sqrt{N}}\big(\sin(2\pi mn/N)\big)_{1\le m,n\le(N-1)/2}.$$

Orthogonality gives $\mathcal{F}_N^2 a=a_{-n}=-a$, so $\mathcal{T}_N^2=I$; no unique positive line is claimed. With

$$\Theta_{a,N}(x)=\sum_{n\in\mathbb{Z}}na_n\mathrm{e}^{-\pi n^2x/N},$$

$$\Lambda_{a,N}(s)=\frac{1}{2}\int_0^\infty\Theta_{a,N}(x)x^{(s+1)/2}\frac{dx}{x},$$

the preceding residue-class proof, replacing 5 by N, gives

$$\Theta_{a,N}(x)=x^{-3/2}\Theta_{\mathcal{T}_N a,N}(1/x),$$

$$\Lambda_{a,N}(s) = \left(\frac{N}{\pi}\right)^{(s+1)/2} \Gamma((s+1)/2) F_a(s) = \Lambda_{\mathcal{T}_N a,N}(1-s).$$

The series expression initially holds for $\Re s > 1$. The integral is entire; the same endpoint bounds and local-uniform differentiation argument apply. This is a finite-Fourier theorem, distinct from an unconstructed N-dependent surface.

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